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From Smart Cities to Manufacturing: A Comprehensive Study on Digital Twin Applications and Barrier

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ABSTRACT

Many sectors are seeing digital twin (DT) as a revolutionary invention, with smart cities, manufacturing, healthcare and transportation being among the most affected. An object or process's DT allows for its simulation via real-time data collection and analysis, thereby making it an exact replica of the original. Based on this capability, data-driven decision-making, system monitoring, product validation and lifecycle management may be realised. DTs, particularly in the manufacturing industry where they improve operational efficiency via predictive maintenance and greater flexibility, have been much more used since Industry 4.0 emerged. DTs find use in aeronautical structural health monitoring and maintenance prediction. Although in smart cities they help urban planning and administration, in healthcare, they provide individualised treatment and patient monitoring. In the energy industry as well DTs are used in turbines to maximise performance and minimise downtime. Though the technique has many advantages, numerous obstacles prevent its general use. Complications in model creation, data integration problems, high deployment costs connected with large sensor networks and computer resources and scalability constraints include among these difficulties. Moreover, the lack of compatibility and consistent standards inhibits more general acceptance. This systematic literature evaluation evaluates enabling technologies and highlights continuous issues that need to be resolved for effective deployment by classifying current research on DT applications across many important industries.

1 | Introduction

Emerging technologies, such as digital twins (DTs), are attracting increasing use in many different sectors. DT technology is a key invention in a variety of sectors, enabling a virtual representation of actual items and systems. Case studies often focus on predictive analytics and lifecycle management [1]. DTs provide complete knowledge of the internal operations of a system, the relationships among its constituents and future behaviour-predicting insights. Smart city, urban development, logistics, healthcare, engineering and automotive stakeholders are just a few of the many that benefit greatly from these practical data. DTs increasing appeal comes from their potential to enhance decision-making, system performance and resource

use. Notwithstanding these developments, the introduction and knowledge of DT technology differ greatly within disciplines and the challenges and constraints of DT application remain unknown [2]. This work offers a thorough study of the DT technology along with an examination of its constraints, trends and difficulties in various spheres, including integration phases. Moreover, providing a particular summary of DT applications with an eye on the advantages, future directions of research and pragmatic implementation advice in addition to technical elements [3].

Industry 4.0 and digitalisation provide many current problem subjects. Apart from cross-company system integration, virtualisation of workpieces, components and assets utilising DTs is

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another main driver of the trend towards better integration [4]. A DT theoretically lets every component and every process virtualized. Integration of both would provide intricate representations of whole systems. However, only then will the digital pictures be able to sustain continuous connection with their actual counterparts and surpass conventional simulation models. Now used across the product life, the DTs (DT) will assist various companies with diverse goals in developing new business models. Not one firm provides a full solution for the DTs as of right now. Different companies use different building methods and use different areas of expertise [5].

The idea behind the DT is to use component connections and performance insights gained from process data. Though it has not yet established itself, it is drawing growing attention from companies and is starting to be important in the industry. One reason is the lack of understanding about the building of a DT. The next part describes the requirements and basic approach for creating a DT [6]. The fundamental framework, which is beneficial for generation, along with many distinct models, is examined. The necessary data for collection, as well as potential means for data transit and storage, are considered. As such, many systems for generating a DT are investigated and compared, and the different approaches for building a DT are clarified. Based on their applications and business strategies, the advantages and disadvantages of certain construction techniques are described [7].

1.1 | Background and Motivation

The background of the paper revolves around the revolutionary potential that digital twin (DT) technology offers in various sectors, such as autonomous vehicles, smart manufacturing, aeronautical, power plant, turbines, virtual cities, healthcare, construction and telecommunications. With industries moving towards digitisation and sustainability under the banner of Industry 5.0, DT has emerged as a vital tool through which it is possible to optimise processes, enhance safety and improve efficiency. With real-time simulations, predictive maintenance capabilities and operational insights, DT technology fills the gaps between physical and virtual systems, leading to smarter decisions and better resource utilisation.

The motivation for this paper comes from the great need for advanced solutions in coping with challenges within different sectors. DT will bring safety as well as an improvement in performance, particularly through virtual testing and predictive analytics, to autonomous vehicles, among many others. In smart manufacturing, DT simplifies operations and improves product quality, whereas in aerospace and power plants, it helps in predictive maintenance and efficiency of operations. It supports the virtual planning of cities in a sustainable urbanisation manner, drives healthcare with real-time patient monitoring capabilities and ensures efficient construction and telecommunication systems. The various applications of DT can highlight universal applicability of DT and its potential to drive innovation, sustainability and resilience in industries.

1.2 | Novelty

Smart cities, businesses, naval, healthcare, energy systems and DT technologies are all being applied here. Research on how DT could enhance biological systems uses real-time data, modelling and predictive analytics. In poorer countries, such as India, inadequate infrastructure, exorbitant expenses and a dearth of skilled professionals impede DT systems. The paper highlights understudied DT uses in public works in smart cities and municipal governance. Main industries, aerospace, healthcare and industrial structures comprise current study results; IoT, artificial intelligence and enormous data are underlined as major forces of development. This paper demonstrates in numerous companies how DT technology influences operational efficiency, sustainability and predictive refurbishment. It also presents important study subjects to increase DT generating acceptance. The research charts developments and difficult scenarios on this theme to direct next projects and assist in removing obstacles to global DT generation.

1.3 | Contributions

- A comprehensive perspective on trending supporting technology and particular tools for DT development: applications. This study seeks to emphasise both for DTs in general and for domain-specific applications the trends in the utilisation of enabling technologies.
- Emphasising the centralised efforts from all sectors worldwide and their various methods; therefore, identifying the overall implementation difficulties both globally and in the Latin American setting.
- Constructing a layered research and assessment of DT applications in many fields concerning the integrity degree.

2 | Related Work

Increased operational efficiency, predictive maintenance and data-driven decision-making follow from the ability of technology to replicate, simulate and monitor actual objects. DT networks (DTNs) use DT concepts in the field of networking to create virtual replicas of network infrastructures, therefore enabling real-time communication, data processing and coevolution between the virtual and physical worlds. This integration is critical for complex networks, such as 6G [8], smart cities and intelligent transportation systems, where DT may improve performance, dependability and flexibility [9]. Achieving sustainable intelligent manufacturing in manufacturing depends in great part on DT-based systems. By means of real-time monitoring and failure prediction, DT maximises manufacturing processes, hence improving quality, efficiency and flexibility and reducing costs. The notion of sustainability is inextricably linked with DT in intelligent manufacturing since it promotes resource efficiency and reduces environmental impact. Sustainable intelligent manufacturing, facilitated by DT, guarantees

that production systems are not only efficient but also flexible to shifting needs and environmental restrictions [10].

As a multidisciplinary technology, DT is rapidly being used in a variety of sectors, including automotive, health and smart cities. Botin-Sanabria et al.'s extensive analysis underlines the obstacles and limits of DT in various disciplines, including as data integration, modelling complexity and system scalability. Despite these obstacles, the potential for DT to transform complex system monitoring, product lifecycle management and data-driven optimisation is obvious. The assessment underlines the need of overcoming these limitations to realise the full potential of DT across businesses [10].

Defining and developing the 'right' DT has been a source of contention, especially in terms of the balance of complexity and utility. Zhang et al. contend that model engineering principles may guide the creation of acceptable DT models, reduce needless complexity while capture the important properties of the actual thing. They propose an evolutionary concurrent modelling method that combines standard modelling and simulation approaches with DT-specific features, resulting in a systematic approach to DT development [11]. The potential of DT technology is also being studied in urban environments, where city DTs (CDTs) provide chances to improve smart city design, operation and administration. Emphasise the early phases of CDT research, emphasising that integrating socioeconomic components and reciprocal interaction between real and virtual city settings are significant obstacles. The progress of CDT technology is seen as a crucial facilitator of smart city development, providing opportunities for better urban administration, sustainability and resource management [12].

DT technology is gaining traction in a variety of areas, including networking, manufacturing and urban management. Its capacity to gen-experiential models of physical phenomena expands the potential for optimisation, real-time monitoring and predictive analytics. However, the effective application of DT confronts many hurdles, including modelling complexity, data integration and the need for multidisciplinary methods to fully realise its promise. As DT research and development proceed, solving these obstacles will be key to realise the technology's revolutionary advantages [13].

2.1 | Research Gaps

The digital twin (DT) idea has received a lot of interest in a variety of fields, including manufacturing, aerospace, healthcare, power systems and smart cities. A thorough analysis of the available literature was undertaken to gain a better understanding of this technology's evolution, applicability and limits. The goal of this section is to recognise research advances, highlight technical improvements and define the remaining gaps that must be solved in order to accelerate DT deployment. The examined studies span a wide range of topics, including predictive maintenance, virtual simulations, IoT integration, data interoperability and real-time system monitoring. Table 1 presents a complete review of the important literature on digital twin technology, including the year of publication, principal

topic of each study and research needs found. This comparative overview contributes to an understanding of the current research landscape and lays the groundwork for future innovations and interdisciplinary collaborations to strengthen DT applications across various sectors.

3 | Methodology

This section covers the methodological approach used to answer significant research issues around digital twin (DT) technology. The research takes a qualitative and analytical approach to investigating DT ideas, difficulties, applications and future trends, with a specific emphasis on the Indian context. The study intends to uncover gaps in DT adoption in industries including as manufacturing, healthcare, smart cities and infrastructure. The following research topics lead this examination and contribute to a better understanding of DT's practical significance and technical progress.

3.1 | Research Questions

- RQ1: what is DT?
- RQ2: where DT can use and wrote applications?
- RQ3: How IOT and AI integrating in autonomous vehicles?
- RQ4: What are the problems with using a DT-based smart city tool in India and other places?
- RQ5: Challenges DT in India?
- RQ6: What are the future trends in how people will use technology that makes things possible?

3.2 | Road Map

In this paper, the Introduction defines digital twins, discusses the study's originality and highlights its significant contributions shown in Figure 1. Related Work provides a complete evaluation of prior studies, publications and current research trends. The Methodology section outlines the study framework, questions and literature route. Digital twin discusses definitions, major components and the role of DT in current enterprises. The Applications section focuses on DT's use in manufacturing, aircraft, power plants and healthcare. Challenges outlines current limits and concerns with India's DT adoption. Finally, the Conclusion presents a summary of the findings, applications and future research directions.

According to the keyword co-occurrence network returned by the VOS viewer, digital twin is the most frequently and central word and it is important to highlight its central role in the present-day research. The clusters surrounding point out major thematic priorities: AI-based modelling and analytics by using machine and deep learning, intelligent automation and optimisation with the use of artificial intelligence and Industry 4.0 innovation and robotic-driven applications. Other clusters include clusters on structural health monitoring, maintenance, transfer learning and cybersecurity. In general, the graphical

TABLE 1 | Research gaps on digital twin.

Ref	Year	Description	Research gaps
[14]	2003	This journal introduces the foundational concept of digital twin by Grieves, setting the stage for future developments in digital modelling.	The paper provides an initial conceptual framework but lacks real-world implementation examples and detailed technical methodologies.
[15]	2012	Published by NASA, this paper defines digital twin technology for aerospace applications, emphasising predictive maintenance and structural integrity monitoring.	Lacks application of DT in nonaerospace sectors; limited discussion on data processing and computational requirements.
[5]	2017	This paper explores the role of digital twins in smart manufacturing, particularly focusing on factory floor simulations and real-time monitoring.	Fails to address the integration of DT with IoT and other Industry 4.0 technologies; minimal discussion on long-term data management.
[16]	2020	Discusses digital twins as 'living models' that continuously adapt to operational changes based on real-time data, with applications in various industries.	Limited in exploring the security and privacy implications of real-time data in digital twin frameworks.
[17]	2020	Reviews the use of digital twins in manufacturing, covering modelling at different scales from atomic structures to complete systems.	Does not address the challenges of cross-platform data integration or the interoperability of digital twin models across industries.
[18]	2016	Provides an in-depth look at the lifecycle management capabilities of digital twins in industrial settings, focusing on predictive maintenance.	Lacks exploration of resource requirements and fails to provide an updated definition of digital twin, which may cause confusion in its scope.
[19]	2020	Focuses on using digital twins for predictive maintenance in aerospace, providing case studies on monitoring structural integrity.	Limited to aerospace; does not explore applications in other industries, such as healthcare or urban planning.
[20]	2021	Analyses the potential of digital twins for urban management, traffic flow optimisation and environmental monitoring.	Gaps in addressing the scalability of digital twins for large-scale urban environments and managing data privacy for citizens.
[21]	2022	This paper investigates the application of digital twins for real-time patient monitoring in healthcare settings, focusing on patient data simulation.	Lacks a comprehensive discussion on the ethical and legal aspects of data sharing and patient privacy in healthcare digital twin applications.
[22]	2023	Provides a summary of the latest trends in digital twin technology, focusing on its applications in Industry 4.0 environments and integration with IoT and AI.	Fails to address challenges in data interoperability and the lack of a standardised framework for integrating DT with other technologies.
[23]	2015	Discusses the application of digital twins in improving production line efficiency in manufacturing.	Limited discussion on cost implications and integration challenges in legacy systems.
[24]	2016	Examines the role of digital twins in predicting equipment failures in power plants, improving uptime.	Focuses on single equipment; lacks discussion on system-level digital twins for entire power plants.
[25]	2017	Covers the integration of IoT with digital twin for real-time monitoring in industrial settings.	Limited analysis of security concerns and data privacy issues in IoT and digital twin integration.
[26]	2018	Investigates the use of digital twins to enhance predictive maintenance in the oil and gas sector.	Does not discuss cost efficiency or the specific challenges of offshore environments.
[27]	2019	Explores digital twin applications within cyber-physical systems in manufacturing, with case studies.	Limited scope on real-time communication challenges in high-speed manufacturing lines.
[28]	2023	Reviews digital twin applications in healthcare for remote patient monitoring, focusing on chronic diseases.	Lacks a discussion on standardisation of patient data formats across platforms.
[29]	2023	Investigates the role of digital twins in robotic automation and real-time control within manufacturing.	Lacks discussion on multirobot coordination and synchronisation challenges.
[30]	2023	Examines the use of digital twins for real-time analytics in energy production and distribution.	Limited to electricity; lacks discussion on the applicability for renewable energy sources.
[27]	2022	Analyses the integration of blockchain with digital twins for secure data management.	Limited exploration of scalability issues and real-world implementation challenges.

illustration highlights how the digital twin research is a multi-disciplinary field, and it connects AI, IoT, robotics and predictive maintenance to promote intelligent systems and smart manufacturing as shown in Figure 2.

4 | Digital Twin

RQ1. What is DT?

The history of DT technology is covered in this part along with some basic ideas and dispelling some common misunderstandings about its identification. DTs first became formalised in the early 2000s. However, changing viewpoints suggest that the technology may have been dreamt of far earlier.

A summary of DT definitions taken from several sources is Table 2. A DT is essentially a real-time digital clone of a physical item that, using data-driven models, not only reflects but also improves the real object.

Growing usage of digital technology has produced transforming DTs—accurate digital copies of physical systems or machinery. DT technology is used widely in many industries and companies to find problems and improve output. Digitalisation is accelerating thanks in great part to information technology innovations, such as cloud computing, big data analytics, artificial intelligence (AI) next generation ITs, such as the Internet of Things (IoT) [38] and data security. The meeting of the actual and virtual worlds has made digitisation the main driver of creativity in all spheres. Four phases constitute the progress of digitalisation indicated in Figure 3: digital enablement,

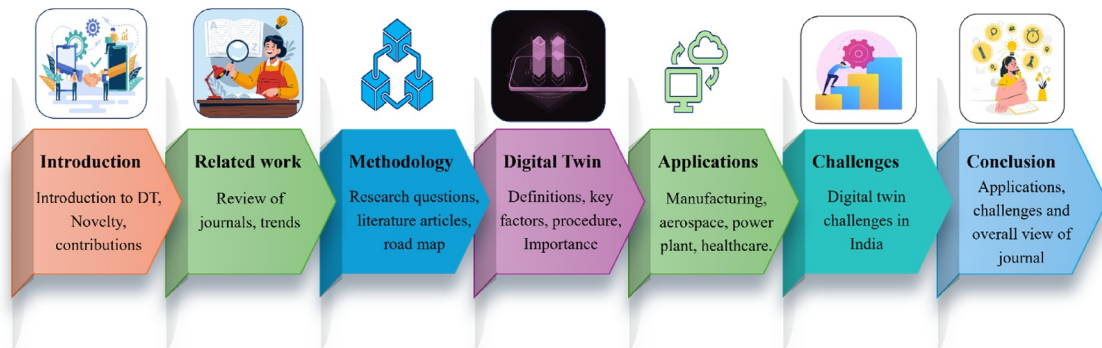


FIGURE 1 | Roadmap of paper.

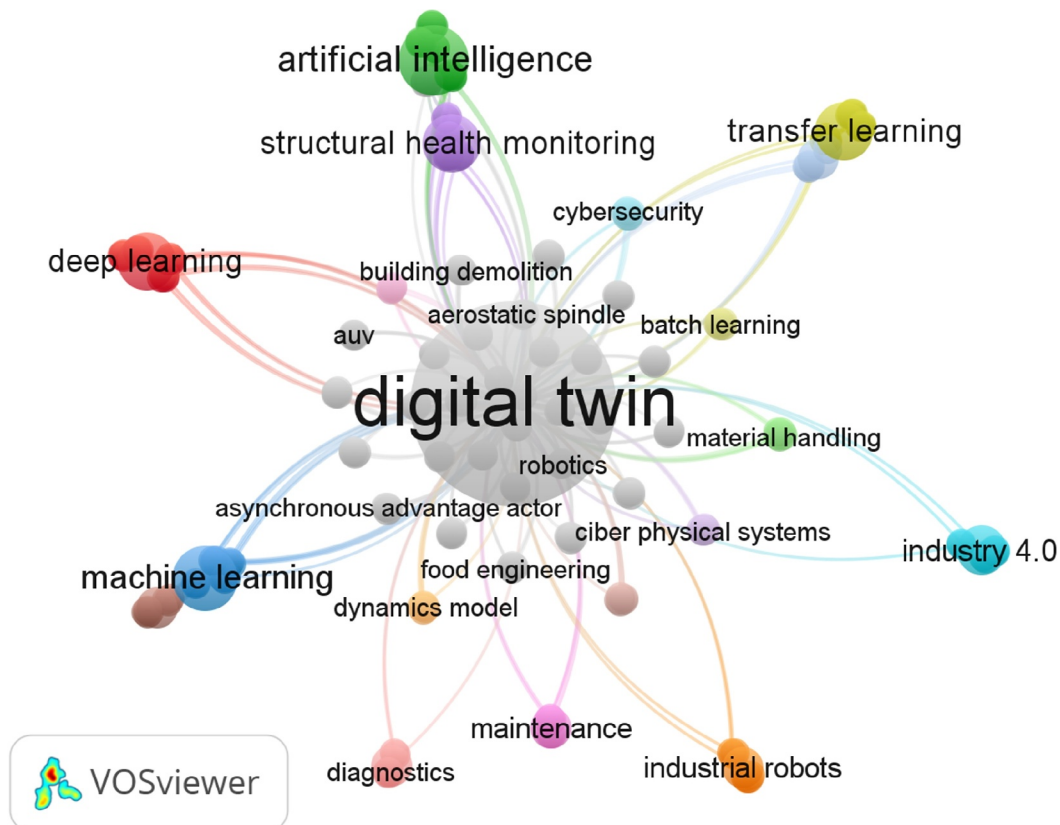
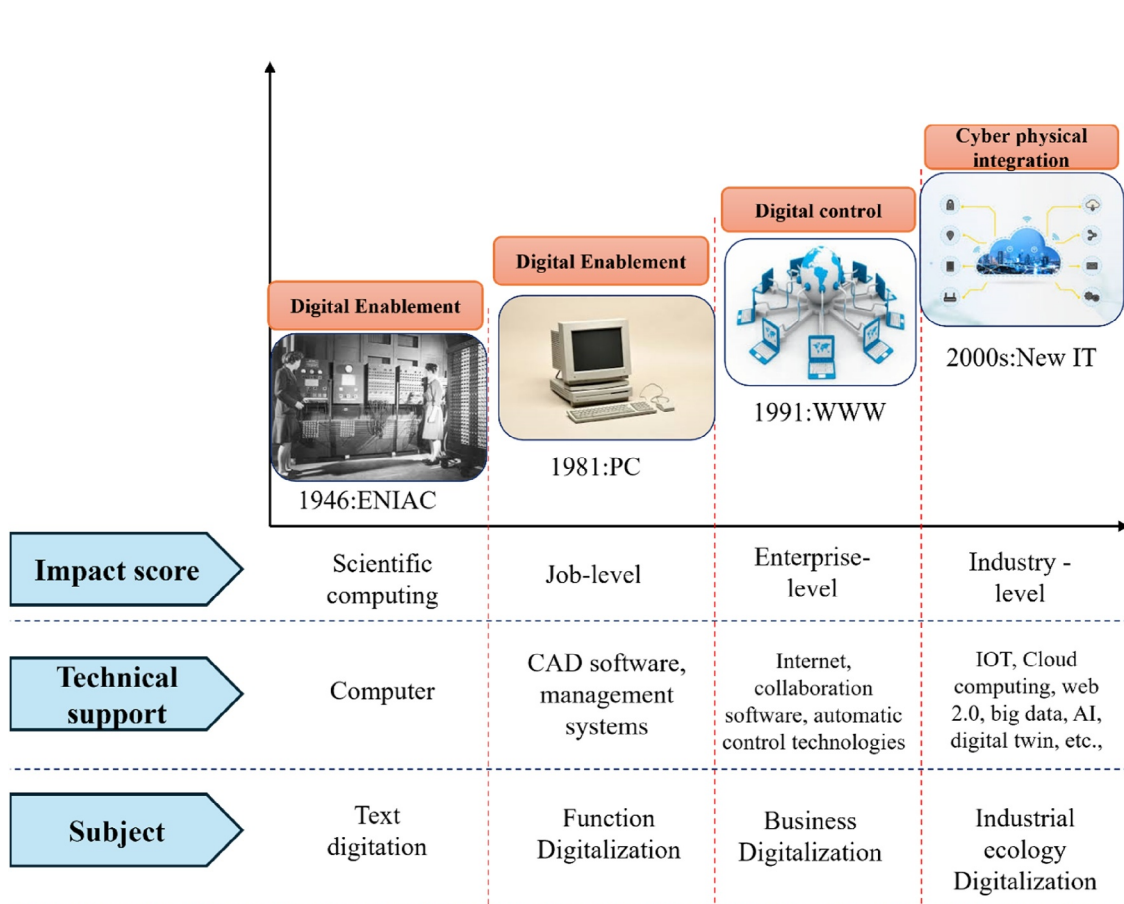


FIGURE 2 | Collection of DTs papers with keywords.

TABLE 2 | Definitions of digital twin.

Ref	Author	Definition
[31]	Chen	Chen said of a DT ‘a digital representation of a physical equipment or system that connects with the operating parts and reflects all functional aspects’.
[32]	Liu et al.	Liu and colleagues describe a DT as ‘live model of the physical asset or system that continuously adjusts to operational changes depending on gathered online data and may project the future of the related physical counterpart’.
[15]	Glaessgen and Stragel	A DT is a probabilistic simulation of a completed vehicle or system modelled by data from sensors, ideal physical models and the history of the associated real thing. It is all-encompassing, multidimensional and multiphysics.
[33]	Abdulmotaleb	Whether biological or not, the term ‘DT’ characterises an electronic copy of any physical object. Integration allows data to go between the virtual and real worlds so virtual objects might live with real ones.
[34]	Lee et al.	Combining data-driven analytical algorithms with already-existing physical principles, a networked simulation of real-world hardware running on a cloud platform models health issues.
[35]	Soderberg et al.	Real-time system optimisation applied using digital models of actual systems.
[36]	Bolton et al.	Real-time data collecting helps one to understand, learn from and reason with the always shifting digital representation of a real-world item or system one.
[37]	Tao et al.	Combining physical, digital and interactive data, the DT shows all elements during a lifespan of a product.

**FIGURE 3** | Evolution of digital paradigm.

digitalisation assistance, digital control and linkage and cyber-physical integration. The term ‘digital enablement’ refers to the preservation of basic characteristics when converting paper documents into digital forms. Only the most important data are first transformed to digital format for storage, processing and transfer uses due to limits of digitalisation technology [39]. The 1980s’ arrival of CAX technologies, such as CAD, CAE and CAM, transformed the view of digitisation and enabled engineers’ computer involvement. Corporate digitalisation was helped by the development of the Internet and improvements in control technology throughout the 1990s, therefore offering companies fresh means of income production and value creation. Emerging technologies are converging the digital and physical domains, producing cyber-physical integration and the digitalisation of industrial ecology from the Internet of Things (IoT), cloud computing, artificial intelligence (AI), big data and the start of the 21st century [40].

4.1 | Cyber-Physical Systems

Cyber-physical systems originated from the proliferation of applications of embedded systems. It was there that the term ‘cyber-physical system’ was first used by Helen Gill at the National Science Foundation (NSF) in 2006 to describe more advanced systems than those that could be characterised using traditional IT frameworks. Since then, CPS has been considered the most important research area requiring investment in the United States. In Germany, a wide recognition exists of how CPS is seen as forming the basis for Industry 4.0, yet their economic potential in redefinition of industrial processes is now being realised. However, research concerning CPS has had a bias on conceptualisation, architecture, technologies and challenges, and practical use in engineering has only begun to emerge recently. Unlike IoT, or sensor technologies, or embedded systems, CPS is more of a foundational category, rather than a specific engineering solutions category, with a bent towards scientific exploration. Therefore, NSF asserts that this research is focused on determining new scientific principles and related technologies. As shown in Figure 4, the shift from traditional IT to advanced IT

has led to the promotion of CPS and DTs in industries. The fact that modern IT is low-cost and has more efficient capabilities has opened the route for the integration of CPS and DTs, placing them as integral parts in smart manufacturing. This, in turn, has the potential to change manufacturing systems as well as business models fundamentally.

CPS is thereby defined as the integration of computational processes with physical systems [41]. On the other hand, a ‘Digital Twin’ (DT) is meant to be a digital counterpart of a physical system whose purpose is real-time optimisation. Two major aspects form both CPS and DT within the manufacturing sector. This consists of the physical and the digital/cyber layer. The former covers the human resources, the machines, materials and even the environmental factors while doing their manufacturing activities, but the latter is associated with all the applications and services through which smart data management, analytics and computation become feasible. These instruments empower the manufacturers to raise the bar of productivity since data are collected, analysed and acted upon in some form. The digital layer is interfering with the physical process and provides data to the physical layer. For example, in Wang et al.’s Wise-Shopfloor, a cyber-physical environment is provided where in situ real-time monitoring and remote control of the manufacturing processes are possible in multiperspective visualisations offered by Java 3D models driven by real sensor data [42].

The heart of CPS is embedding computation and communication capabilities into physical systems in close interaction with physical processes. The core 3C concept of CPS is a combination of computing, communication and control to allow real-time sensing, dynamic control and information services for complex systems. Although CPS emphasises computation and communication of the world of cyber, it extends the precision and functionality associated with physical activities. With three-tier, five-tier or even service-oriented architectures for CPS, most architectures have concentrated on controlling aspects rather than mirrored ones. Feedback is very crucial in any kind of interaction and cooperation between digital and the physical layer of both DPS and DTs. However, the association of cyber

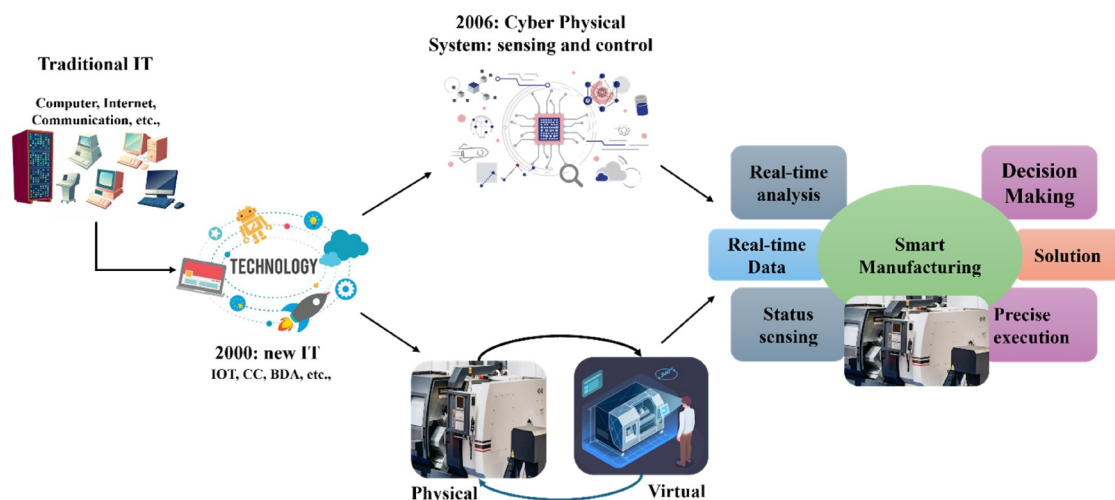


FIGURE 4 | Cyber physical system.

and physical components of CPS may be one to many because several physical elements can be influenced by one computational system. That makes it possible for more complex and diversified physical systems to be controlled with increased efficiency by CPS [43].

4.2 | Key Factors

Thanks to DT's meteoric development, new channels and platforms have opened, allowing for the simplest possible implementation of products and services. When it comes to creating digital or virtual twins of physical assets, DT is considered a front-runner among digital platforms. To analyse data and keep an eye on systems, DT functions depend on the bridge between the actual and virtual worlds. Highly effective communication channels that make use of real-time updates enable the link between the actual object and the virtual representation. Combining sensor networks with cloud computing and big data analytics helps the Internet of Things (IoT) to become a potent communication tool. DT's functioning is mostly based on data and information sent between the two ends [44]. Over the lifetime of a product, it makes data flow between a physical and digital asset flawless possible. Through information and real-time data collecting, a DT can constantly adapt to operational changes and replicate a physical asset, therefore offering forecasts about how those changes will impact the asset going forward. By using this, it may move from reactive to predictive maintenance techniques, which include forecasting when the physical asset will need repair [45]. The DT systems-of-systems integration includes embedded sensors, wireless sensor networks, digitalised building life-cycle data and systems and cloud services and data providers. Real-time data allow a DT to adapt to environmental or operational changes and deliver the optimum outcomes. One method is using sensors to accurately synchronise the DT with the actual asset over the internet. Data are collected in real time by sensors updating the virtual model. DT comprises the physical product, the digital product and the communication channel that connects them. A DT requires analytics throughout to integrate a product's virtual and physical components. Described various DT traits, including: A DT is a health, repair history, performance and other vital properties of a physical structure [46]. A DT can be used to achieve reliability and efficiency by creating preventative maintenance schedules by constantly checking these parameters and studying the historical data. It gauges the performance of the asset through its digital representation, determines its present condition and provides the future operation states and maintenance needs. This will allow facility managers and developers to track the performance of the system in real time and implement required changes to adjust to changing needs. In addition, the digital thread connectivity also enables the tracing of all life cycle stages of the physical asset. Predictive analytics improve this capability by improving assumptions and predicting future system behaviour. Also, it allows remote maintenance because the DT can diagnose and fix the problems with faulty equipment remotely. A DT can even replicate ageing process of a real system by mimicking the operational data and maintenance records, which will be very dangerous in offering meaningful

information on the long-term performance and sustainability of that system.

4.3 | Procedures for Implementing DT Technology

In its operations, DT adheres to the guidelines laid forth by the IoT. The level of information needed to create the model and provide continuing input, together with the amount of IoT adoption, will determine the implementation. The amount of data used in the virtual model is determined by the goal of the DT application. Finding the right balance between complexity and level of detail is an important first step when using DT technology [47]. According to Deloitte (2017), there is a six-step process for DT technology implementation:

- a. Creating possibilities: This step involves thinking about possible possibilities and assessing them based on established criteria to choose the best course of action.
- b. Finding the Right Process: Considering aspects, including organisational, business, and operational change management, this entails identifying the digital transformation Configuration that provides the most value and has the best chance of success.
- c. Put the DT into action: A trial programme is launched using iterative and agile cycles to speed up learning, proactively reduce risk and ensure ROI. The first outlay of capital is maximised.
- d. Make the process industrialised: Using existing tools, procedures and playbooks, the piloted DT scenario is commercialised to ensure it can be used in the appropriate domains.
- e. Broaden the twin's reach: The piloting phase's values, along with the tools, techniques and playbooks, are shared with stakeholders and the rest of the enterprise to make sure the DT meets their needs and improves it.
- f. Keep an eye on things: To make sure the results are as good as they can be, measuring and tracking the how solutions are working to determine how much value they added [48].

4.4 | DT Importance in Industry 4.0

DTs, computer simulations of real-world assets, processes and systems are crucial in Industry 4.0. By using real-time sensor data, these digital simulations allow firms to assess, monitor and improve processes with an unparalleled level of accuracy. DTs allow businesses to improve decision-making, simplify maintenance, increase workflow efficiency and anticipate equipment breakdowns. This technology enables the construction of smart factories, warehouses and logistics systems, significantly enhancing visibility and management. As a result, DTs drive significant improvements in performance and innovation within Industry 4.0 shown in Figure 5.

By creating virtual copies of real-time monitored, simulated and operationally optimised actual assets, systems and processes, DTs are revolutionising manufacturing. These digital models improve operational efficiency by helping manufacturers to spot possible difficulties, streamline processes and forecast equipment failures, resulting in less downtime and lower maintenance costs. Furthermore, DTs enhance product quality by enabling virtual testing and design improvement, thereby reducing mistakes and waste. They help shorten time-to-market by reducing the need for physical prototypes and enabling quick changes throughout manufacturing. Furthermore, the data insights created by DTs facilitate improved decision-making, allowing enterprises to adapt quickly to market needs and operational issues. DTs, with their capacity to optimise resource utilisation and encourage sustainable practices, are emerging as a vital instrument for improving smart manufacturing and driving the transition to Industry 4.0.

4.5 | DT Importance in Industry 5.0

The industrial sector has evolved rapidly as producers all around struggle to boost output. Germany first Industry 4.0 is based on a tiered industrial development progression shown in Figure 6 from Industry 4.0 into Industry 5.0, initially proposed Industry 4.0 in 2011 to boost industrial competitiveness. Emphasising sustainability, DTs, human-centricity and resilience among other key goals of Industry 5.0, Industry 4.0's qualities are also contrasted with Industry 5.0's emphasis on decision-making and human robotics cooperation. It also highlights possible Industry 5.0 uses including supply chain management, DTs, healthcare and Society 5.0. Industry 5.0 should also concentrate on building a framework using semantic integration and effective use of big data that maximises processes [49]. Industry 5.0 seeks to improve the part industry plays in helping society, even if it is mostly a vision for developed countries as current issues in efficiency and production

limit its possibilities. Although many businesses still have Industry 5.0 as a future-oriented target, the full potential of Industry 4.0 still must be achieved to reach leadership in industrial innovation and better forecast manufacturing operations [50].

Modern industrial output depends much on the deployment of emerging technologies. DTs technology integrates transdisciplinary simulations at several levels using data from operational history, sensor updates and physical models. DT uses simulation technology to gather data from several physical models, therefore producing virtual reproductions of real-world objects. DT may thus project the physical environment via data accumulation and artificial intelligence feedback by monitoring digital representations and operational measures in real time. Especially, research shows that 85% of Internet of Things (IoT)-based devices in information security employ DT. Consequently, the use of DT in smart city development has become more and more important in research, particularly in relation to urban infrastructure, a fundamental part of IoT systems. Many times, modern building methods overlook the changing demands of society [51].

DT technology started in industrial production. Using 3D simulations, DT can see the real-time condition of a production system. Although in the production process, DT may mimic equipment operation and change parameters; during product research and development, DT provides virtual modelling and verification via simulation tests. This enables producers to lower development and manufacturing risks, increase product dependability and availability and thus enhance their quality. In the maintenance phase, DT is also rather important as the constant gathering and analysis of operational data might assist to forecast the best moment for repair. Establishing maintenance cycles and a basis for determining possible failure spots and probability depends on this predictive power. Industrial executives in specialised sectors have been rather interested in DT as it increases operational efficiency and reduces costs.

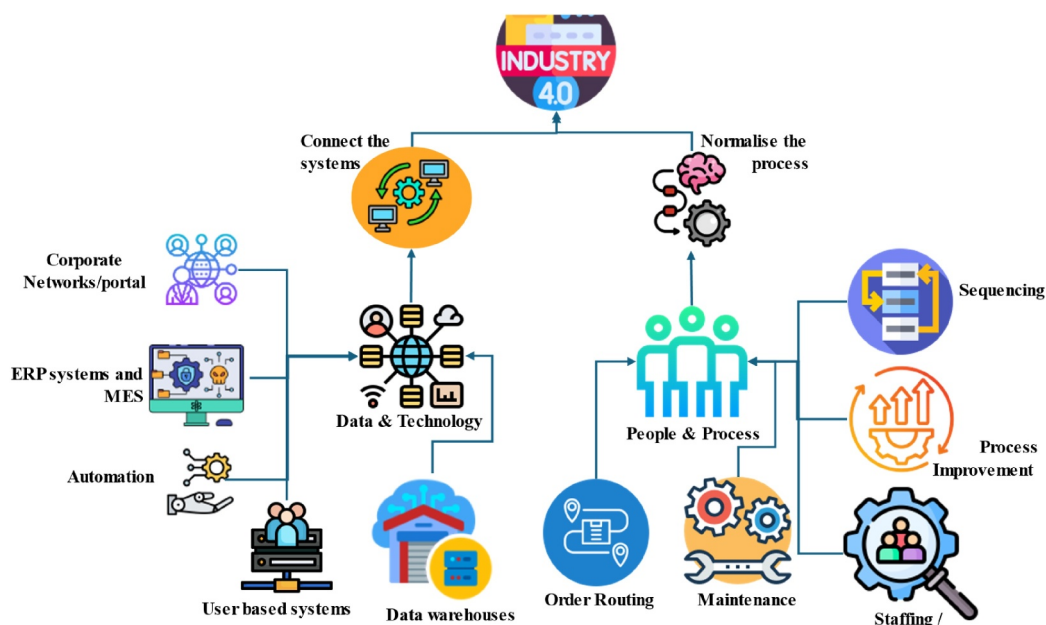


FIGURE 5 | Role of Industry 4.0.

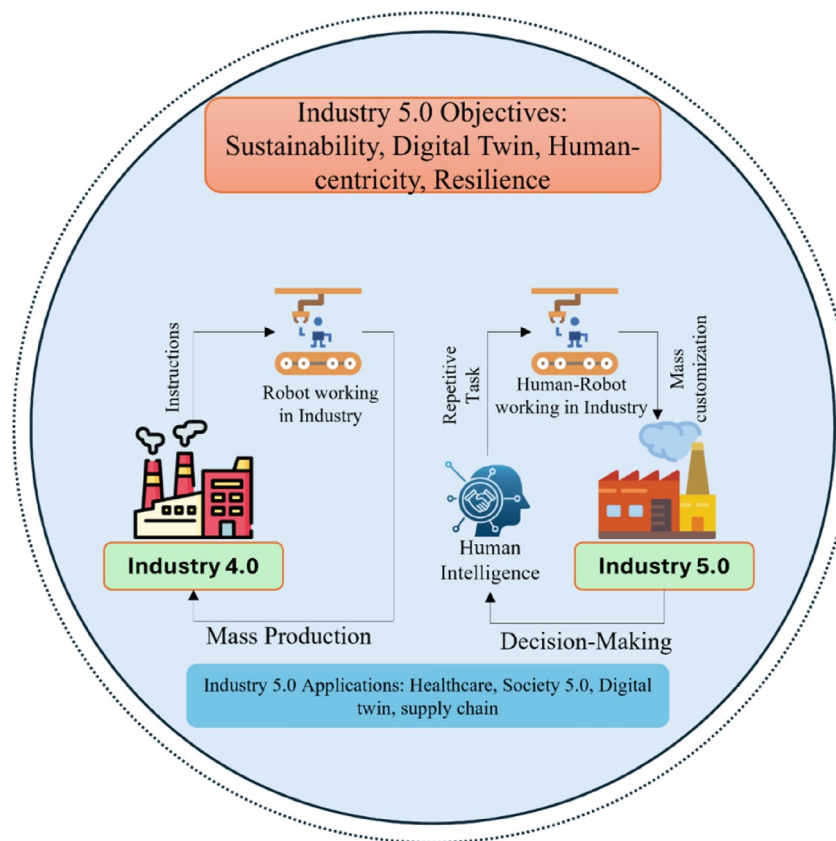


FIGURE 6 | Concept of Industry 5.0.

Research on the use of DT in industrial sectors presently largely focuses on troubleshooting and testing within virtual environments, with an eye towards perfect operational results for equipment and other linked systems. Still missing is thorough investigation on the overall development and use of DT in industrial production. This was addressed by extensive research on the use of DT inside the Fifth Industrial Revolution [52].

5 | Digital Twin Case Studies

5.1 | Manufacturing

5.1.1 | Manufacturing in India

Manufacturing plays a pivotal role in the Indian economy, significantly enhancing the gross domestic product (GDP) and employment opportunities domestically. The COVID-19 epidemic has caused significant interruptions even although the industry has had consistent increase over the years with an average growth rate of 5.79% from 2006 to 2024. Lockdowns implemented in April 2020 cause a terrible drop in industrial production, reaching historically low levels of -66.60% as shown in Figure 7. Nevertheless, in April 2021, the sector saw a remarkable 196% rebound, achieving its highest level of rapid recovery ever recorded [53]. This demonstrates the economy's resilience and the effectiveness of the reopening measures. Subsequently, the manufacturing sector expansion has been consistent, with a modest 4.6% rise in July 2024 compared to the previous year. In conjunction with an increase in foreign direct

investment, initiatives, such as 'Make in India', aimed at converting India into a global manufacturing powerhouse, have bolstered the nation's manufacturing sector. Even in the face of challenges, such as infrastructure limitations and supply chain disruptions, the sector remains a key driver of national economic growth. Local considerations and global economic situations may influence the industry's pace but forecasts indicate a sustained growth trajectory in the future [54].

5.1.2 | History

Although there have only been five industrial revolutions so far, manufacturing automation depends especially on the third and fifth ones. Industry 3.0 is driving automation in the manufacturing-business to raise whereas industry 5.0 will need human interaction with co-bots as they collaborate in the manufacturing industry in the future as shown in Figure 8.

5.1.3 | Smart Manufacturing With DT

The image stresses the role of DT predictable in driving Industry 4.0 by demonstrating the interconnection of physical and virtual production infrastructures from Middlesex University and Siemens. The image shows how sophisticated models and machine learning can accurately replicate real-world goods, such as cyber-physical systems, in virtual surroundings [55, 56]. This generates a DT, which improves the design, specification and validation processes. This continuous feedback loop between

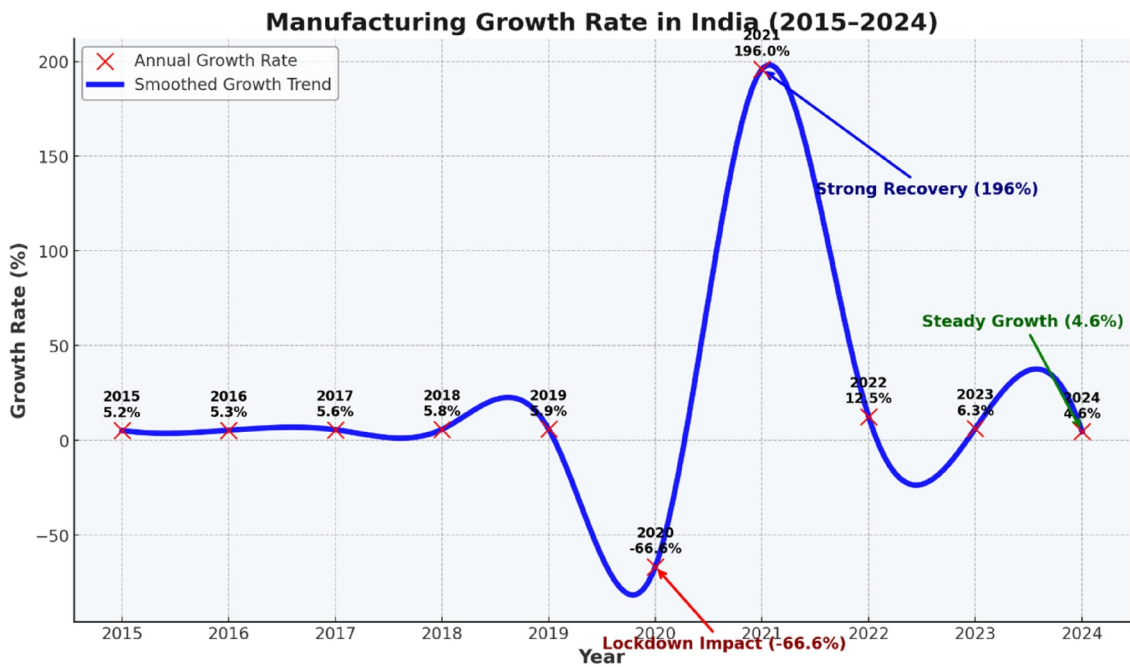


FIGURE 7 | Manufacturing in India.

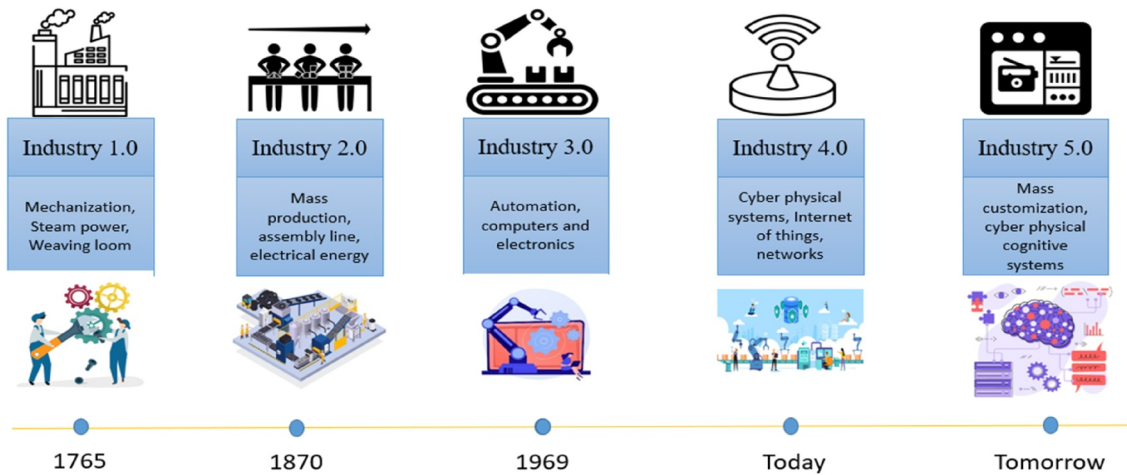


FIGURE 8 | Industrial revolutions.

actual and virtual goods enables real-time data transfer, resulting in predictive maintenance and diagnostics [57]. The DT improves system performance and production by leveraging real-time operational data insights shown in Figure 9.

The DT enhances production and system performance by leveraging real-time operational data insights. Platforms, such as Siemens MindSphere, collect data from actual products and deliver actionable insights, hence improving automation and operational efficiency [59]. The relationship between physical systems and their digital equivalents reinforces the continuous improvement cycle, resulting in smarter operations, more efficient processes and reduced downtime. Finally, the graphic underscores the importance of DT technology, which is backed up by real-time analytics, automation and predictive tools, in

developing intelligent and sustainable production systems in the Industry 4.0 era [10].

Imagine a plant where real-time production optimisation and maintenance required prediction between communicating equipment. DT-based smart manufacturers see this as their ideal. DTs help manufacturers to get hitherto unheard-of insights into their operations by building a virtual counterpart of the actual plant, therefore fostering more efficiency, lower downtime and better product quality. A DT based smart factory is fundamentally based on the Internet of Things (IoT). In the physical factory, Actuators and sensors gather and forward data to the DT, therefore forming an ongoing feedback loop. Advanced analytics and simulations made possible by this real-time data let manufacturers find bottlenecks, streamline

operations and foretell possible problems before they start. DT-based smart factories provide rather enormous advantages. From better quality control and lower costs to increased decision-making and predictive maintenance, these technologies are transforming the production of goods. Years to come should show even more creative uses and advantages as DT technology develops shown in Figure 10.

RQ2. Where DT can use and wrote applications?

5.2 | Aero Space

DT technology has been successfully used in a variety of sectors, including aerospace, where it has shown great promise for improving structural integrity and maintenance operations. DT has been used at the Structural Sciences Centre of the US Air Force Research Laboratory to create high-fidelity flight models combining virtual simulations with real-world data to precisely estimate aircraft lifetime of fa-

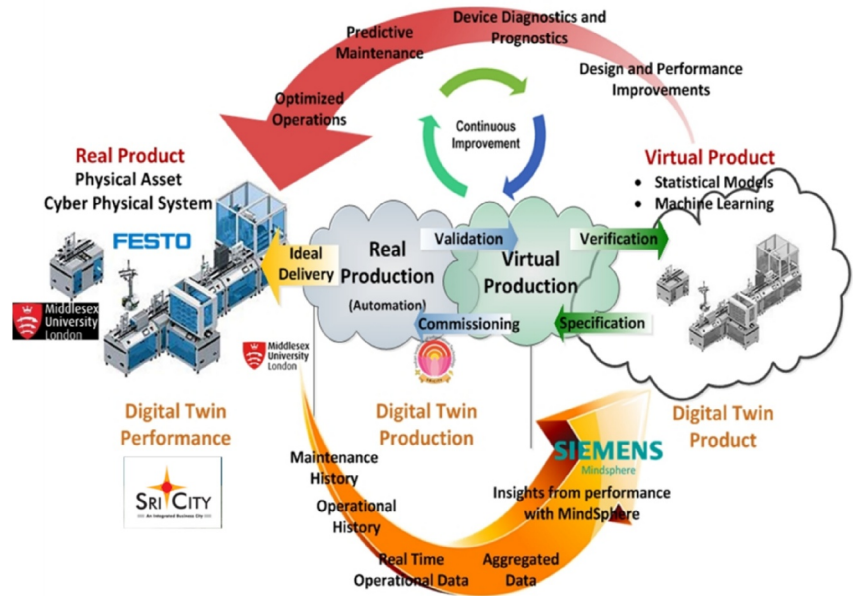


FIGURE 9 | Proposed DT framework [58].

Digital Twin based Smart factory – a CPS approach

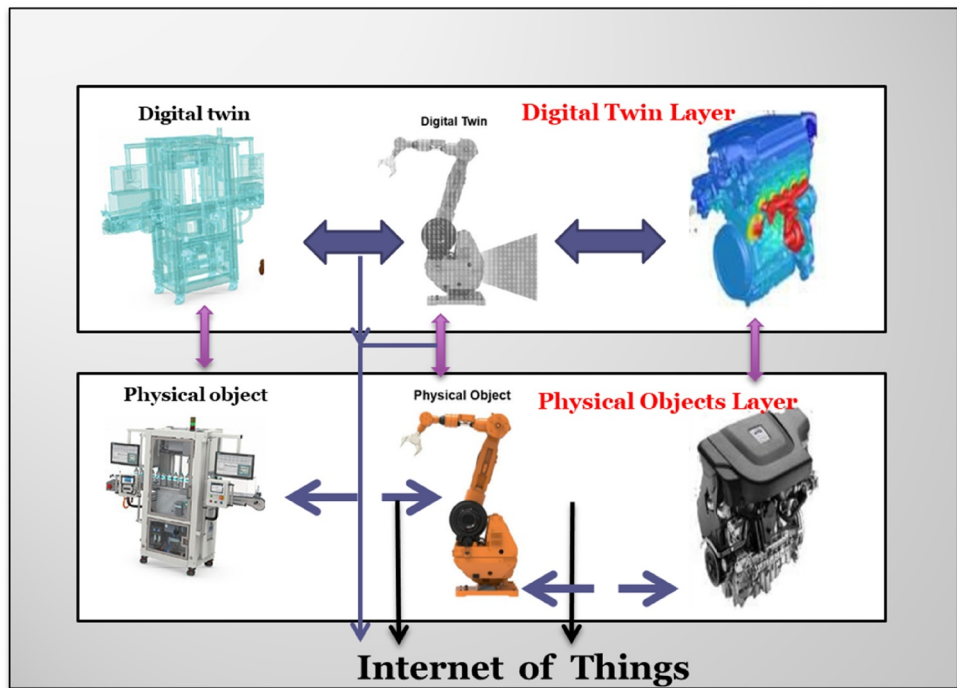


FIGURE 10 | Different phases of the DT model.

tigue. More accurate structural wear and tear predictions made possible by this breakthrough helps to create more successful maintenance plans. Previously, especially during reengineering activities, DT was seen as a tool for assessing aircraft structural life. Many times, constructed for weight economy, aircraft are prone to fatigue failure. Calculating operating life to solve this issue depends on fatigue analysis using molecular-level aircraft geometry [60].

Nevertheless, to completely explore these molecular structures, standard methods often need for destructive testing. Lead by Airbus, the Clean Sky 2 research initiative in Europe is creating new sensor systems to track aircraft structural condition, most importantly landing gear. The large aerospace and military firm TEST-FUCHS also employs DT-based technology for their test systems and ground support equipment. DT is used in many contexts than only aircraft. Using networked technologies, the US Department of Defence's Digital Engineering initiative seeks to integrate systems and activities. DT has also been used to health management and equipment prognosis, therefore enhancing the capacity to foresee and prevent issues in many other spheres [61]. DT has been used in aviation maintenance to control structural

health, therefore enabling exact damage prediction including the position, magnitude and direction of possible failure. To find and repair damage, DT models-including the Airframe DT (ADT-assess flying conditions in real time). For structural defects in aeroplane wings, for instance, the method has been used to track them and provide useful data for next maintenance. Although DT is utilised largely in aviation, its increasing popularity in this sector offers strong proof for its use in other sectors including building [62, 63]. The graphic that has been supplied provides an overview of a conceptual framework that addresses the implementation of DTs in the aerospace sector, more especially for the manufacture and operations of aeroplanes shown in Figure 11. This article explains how a DT, which is a virtual counterpart of a real aircraft, may be used to improve different elements of the aircraft's lifespan, including the design and development of the aircraft as well as manufacturing and operations management. Engineers can find areas for improvement by evaluating and modelling data from many phases of the aircraft's life cycle. These areas include improving design parameters, manufacturing processes and operating settings. Therefore, this eventually results in increased efficiency, decreased expenses and an overall improvement in the performance of the aircraft.

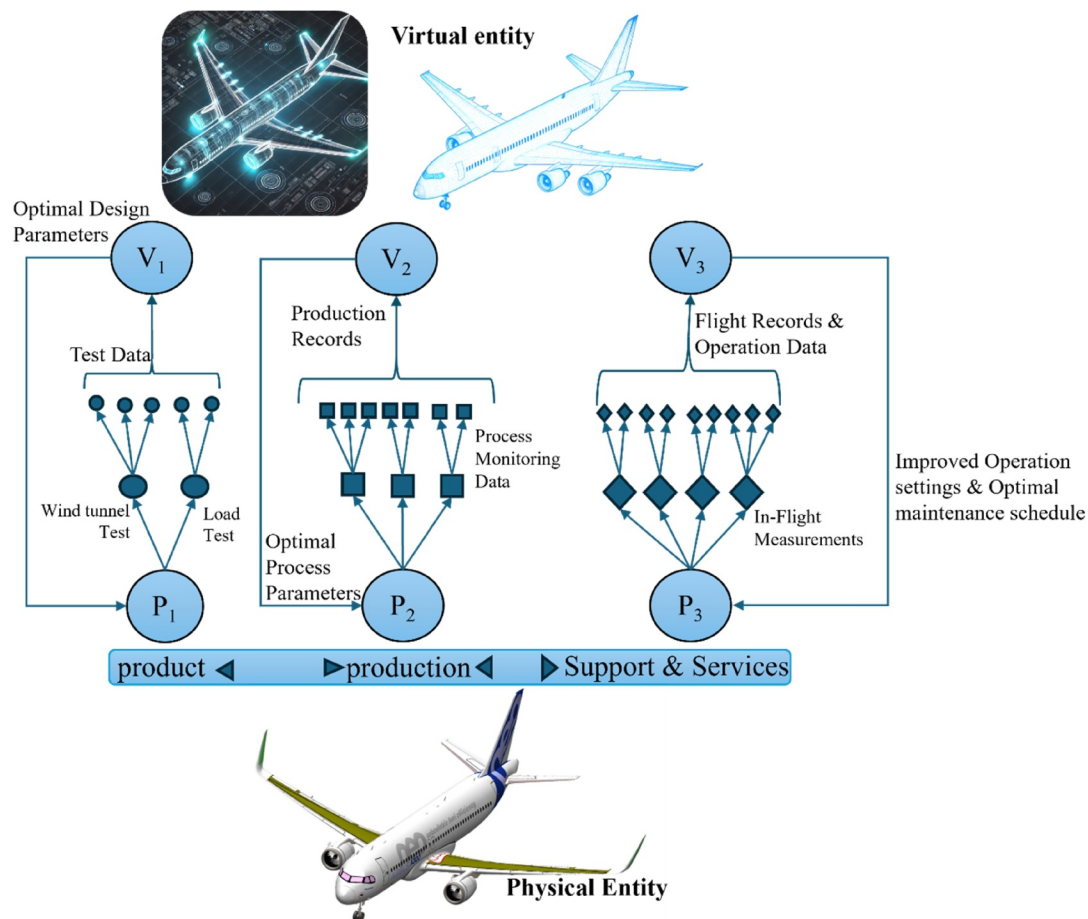


FIGURE 11 | DT of aeroplane.

5.3 | DT in Autonomous Vehicles With IOT

RQ3. How IOT and AI integrating in autonomous vehicles?

In the automotive industry, a DT is the integration of an electronic equivalent of a real car into the manufacturing process. On the left, although driving in the real world, the physical automobile gathers data from all its numerous components, sensors and surroundings. Access to this real-time data are critical for building and maintaining the DT, a mirror of the physical vehicle's real-time performance and condition as shown in Figure 12. Real-time automotive data mean that the computer twin a computer replica of the car may alter in real time. This approach lets engineers duplicate the performance of the car, test it in numerous settings and project issues even without operating it. Given constant information flow between the virtual and real cars, the DT might allow real-time monitoring and optimisation. Data flow is data arrows are important for this system.

The DT tracks events, finds inefficiencies or abnormalities and assesses performance using vehicle data. Though the story still develops, the DT created from using artificial intelligence to the data ends. The feedback arrow illustrates the feedback loop, thereby stressing how the real car may learn concepts and ideas from its digital copy. The vehicle may increase its real-time efficiency by means of engine performance optimisation, predictive maintenance or operational process changes applying this feedback system. Artificial intelligence can predict vehicle faults by analysing intricate data patterns, therefore enhancing driving safety and efficiency [64].

Advancement of autonomous driving systems rely mostly on the integration of artificial intelligence (AI) with Internet of Things (IoT) data so that they may make real-time decisions enhancing vehicle performance and safety. IoT devices incorporated in autonomous vehicles, such sensors, cameras and communication systems collect constantly massive amounts of data on the surroundings, traffic conditions, weather and internal operations in this framework [65]. AI-powered analytics can manage this massive volume of IoT data in real-time especially using machine

learning techniques. By use of sensor inputs, artificial intelligence can foresee possible hazards, optimise navigation courses and change driving behaviour based about affairs. For example, AI may advise the autonomous system to slow down or change lanes after seeing dangers such as pedestrians or other automobiles instantly. IoT device data may similarly alert the system to unanticipated weather changes, therefore directing speed or vehicle handling for optimal safety. This perfect marriage of IoT data and artificial intelligence-driven decision-making also enables predictive analytics, where the AI system projects future events based on historical and real-time data. Predictive maintenance reduces the possibility of unexpected breakdowns by artificial intelligence forecasting when a part of the vehicle might fail using data from IoT sensors. Moreover, improving vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication is artificial intelligence models learnt on IoT data. These technologies may enhance traffic flow, reduce congestion and prevent accidents by allowing autonomous automobiles engage with smart city infrastructure in real time and with one other. IoT combined with artificial intelligence in autonomous driving systems is essentially a powerful combination that increases situational awareness, enables quick and precise decision-making and produces a more efficient and safer driving experience.

5.4 | Power Plant

Following trends of global decarbonisation, decentralisation and digitalisation, the energy sector is seeing major changes. Over the last several decades, big data and artificial intelligence technologies have emerged and been used quickly, making the power producing industry smarter and able to monitor and modify significant assets in real time. Using physical and data-driven models, current hot subject DTs either replicate or predict system status. DTs are fit for many power generating assets. Their objective is to find anomalies or make maintenance decisions by means of system changes detection. Energy revolution calls for DTs more and more. The power generation-specific DT technologies are reviewed in this paper. Lifetime, complexity and DT model type defined the classifications for DT studies for the combined cycle gas turbine, wind turbine, solar and

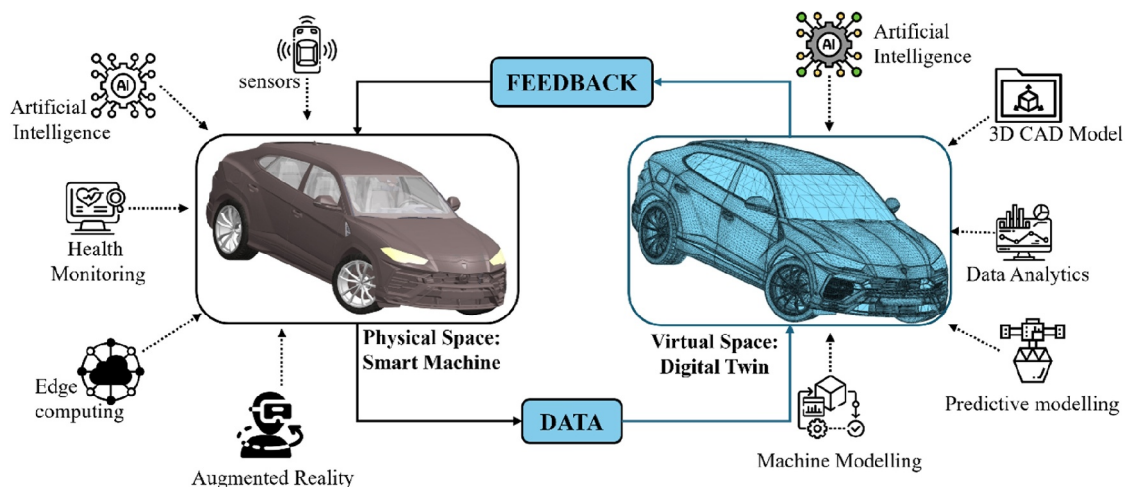


FIGURE 12 | Integration of IOT and AI.

nuclear power plants; their characteristics and constraints were investigated. The goal is to provide readers a condensed summary of the many use cases they can find useful for their job. The paper also addresses reliability, efficiency, availability, environmental problems and future directions of study [66].

The worldwide shift from fossil fuels to low-carbon energy sources is propelling the development of renewable energy, especially wind and solar power. Consequently, coal power stations, particularly in developing countries, are increasingly required to provide peak shaving services to stabilise the fluctuating output of renewable energy sources. Optimising the operation of thermal power plants is crucial for reducing carbon emissions and enhancing efficiency. Digitisation has become a crucial technique for optimisation, with 'smart power plants' significantly contributing to improved dependability, cost efficiency and operational flexibility. The DT approach, the actual power plant to its virtual equivalent, is fundamental to smart power plants. DT technology enables real-time oversight and regulation, enhancing the adaptability and efficiency of thermal units. DT modelling integrates data-driven techniques with physical equipment processes, allowing precise simulations and forecasts. The DT model for thermal power plants enables real-time monitoring of performance, economic and safety parameters, allowing facilities to rapidly adjust to fluctuating loads and enhance energy efficiency [67].

The digital transformation development process for thermal power plants generally consists of three phases: big data analysis, digital modelling and operational optimisation. In the preliminary stage, extensive data are examined to improve plant efficiency as shown in several studies that use distributed computing and data-driven modelling. The second step is developing dynamic simulation models to evaluate operational flexibility and performance. In the concluding phase, operational optimisation techniques are used to enhance energy efficiency and refine operations, including live steam pressure optimisation and cold end optimisation. This research introduces an innovative DT paradigm for steam turbine systems (STS), integrating mechanism-driven and data-driven methodologies. The model delivers precise simulations of critical factors affecting STS energy efficiency and makes recommendations for enhancing real-time operations, resulting in improved overall performance.

5.4.1 | Turbine

A steam turbine is used to demonstrate the notion of a DT, which is a situation in which a virtual representation correlates to a

physical thing via the use of a digital representation. By collecting, analysing and mapping data from the actual thing, it is possible to generate a digital duplicate of the original. The subsequent uses of this virtual reality include, but are not limited to, the modelling of data, the storing of data, the simulation of models and the management of operational processes. Through the process of imitating the behaviour of a real thing in a virtual environment, engineers can evaluate its performance, pinpoint existing problems and improve its overall functionality as shown in Figure 13.

5.5 | Virtual City

RQ4. *What are the problems with using a DT-based smart city tool in India and other places?*

Figure 14 illustrates the notion of a DT, a virtual representation of a real thing. In urban planning, a DT signifies a metropolis, enabling real-time monitoring, analysis and simulation. Data obtained from sensors and other devices are integrated into the DT, forming a complete digital picture. City planners may learn a lot about things, such as traffic patterns, energy use and environmental effect, by studying this data using powerful analytics tools. More effective and environmentally friendly city planning is possible thanks to these findings, which guide decision-making. For example, DTs may be used to enhance transportation networks, forecast the effect of natural catastrophes and uncover energy saving options [68].

5.6 | Healthcare

The literature currently covers medical DT implementations. Fitness, viral infection simulations, smart city well-being, remote surgery and healthcare management are among its many uses [69]. DT technology may aid healthcare administration by using AI, data science and DL to give better faster treatment. Similar technology has been used to produce vital sign monitoring apps, BCIs, nutrition tracking apps and liver and heart disease diagnosis apps. Its main challenges as a human DT (HDT) conceptual paradigm are security and social ethics. It uses wearable sensors, mobile phones and medical data to replicate a person's body in a cyber-physical environment and update its content via online services. To contextualise the patient's health situation, many tools analyse the data. Other factors include contact with others and the environment. HDT theory states that a birth index feeds biological information into the HDT as people develop [21, 70].

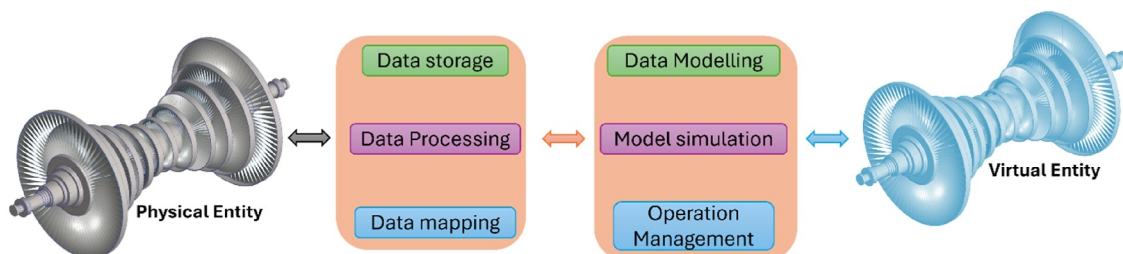


FIGURE 13 | DT of turbine.

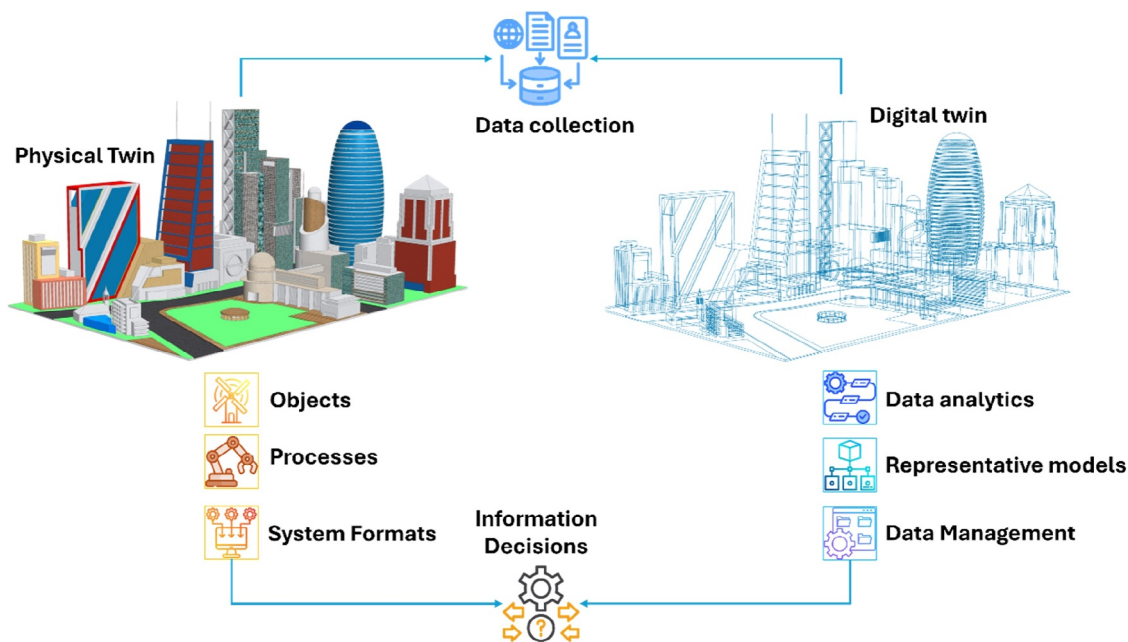


FIGURE 14 | Virtual city.

People are more intricate than industrial processes, thus thoughts, reactions and behaviour may be irregular, even if certain characteristics may be monitored to infer dangers. This makes the HDT problematic. People as models are not independent. Inheritance, genetics and culture may affect human traits. Heartbeat, steps, physical activity, sleep duration and handwritten notes on meals, performance and mood were recorded. The Smurfit app collects data, sends it to predictive models and shows trainers and coaches the model outputs in an easy-to-use graphic for corrections, monitoring and suggestions. ML and DL algorithms may patch gaps or manually modify input data by picking reliable predictors that include previous and future values. The processing time needed to construct distinct models increases with data amount, making this kind of DT harder. As new data becomes available, incremental learning is suggested for 'on the run' learning [71].

Recent research describes a DT that mimics human viral infection reactions. Each subcellular, multicellular, tissue, organ and whole-body model represent viral infection parameters at those scales. The experimental component of this DT is crucial since sensor and medical record data is needed to make reliable simulation predictions. The authors propose a cooperative city DT that provides municipal crisis management information. Another Remote Sens. initiative used a city-scale platform for engaging DTs to make data-driven decisions and develop inclusive and effective COVID-19 control methods. Various drugs, poorly understood biological phenomena such as bodily infections and the immune response to viral viruses are among the major challenges for this kind of DT. Thus, expert validation is crucial to model reliability [72].

An ISO/IEEE 11073 (X73)-standardised well-being DT is described. This work follows worldwide personal health device data collection guidelines. A proprietary system used shoe insole sensors to estimate gait forces in real time to detect whether participants were walking, jogging or sitting. The authors

suggested using visualisations in a smart city to improve citizens' healthcare. Analysing all patient physiological parameters in real time enables the system to send alerts to cloud servers, where medical experts and organisations may take preventive or emergency action. Figure 15 shows an acceptable HDT construction [73, 74].

5.7 | Construction

By providing virtual copies of real-world assets, DTs are revolutionising the construction industry. From concept and building to operation and maintenance, these virtual models provide a complete picture of the project life. Combining data from several sources, including sensors, BIM models and real-time monitoring systems, DTs provide a dynamic and current picture of the physical asset [75]. DTs provide real-time monitoring of construction progress, allowing for early detection of possible difficulties and better decision-making. By comparing as-built models to design blueprints, DTs assist construction projects satisfy quality requirements and save costly rework. DTs can detect possible safety risks and monitor worker movements, lowering the likelihood of accidents and injuries. DTs give significant information about the functioning of buildings and infrastructure, allowing for preemptive maintenance and repairs.

Although DTs have several benefits, their application needs careful planning. Data quality, connection with current systems and the establishment of defined protocols are all significant challenges. Even more creative applications of DTs in the building industry should be expected as technology develops, producing more solid environmentally friendly projects. DTs might help building design and lifetimes management in Figure 16. Should it be required, a 'DT' is a perfect digital replica of a real-world physical item meant to replace the original one.

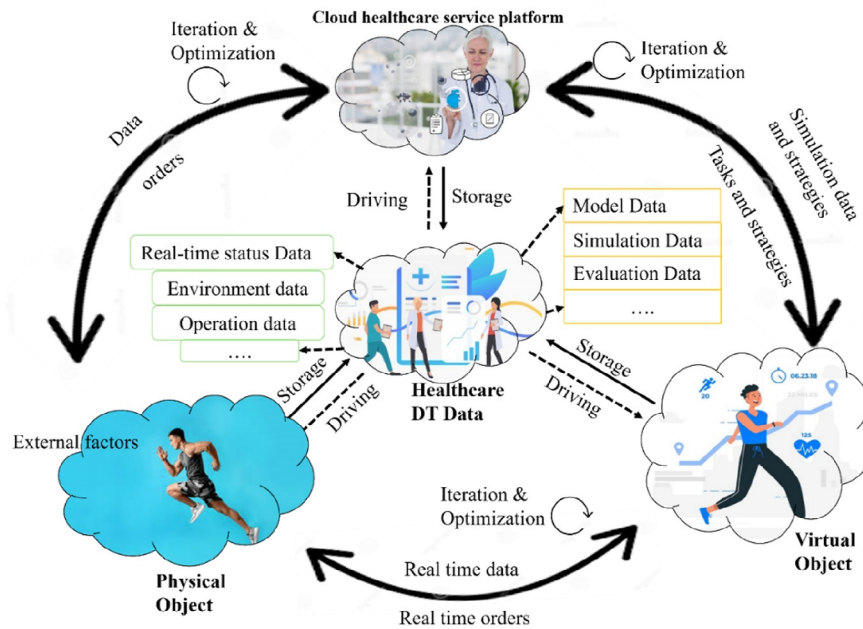


FIGURE 15 | The framework for DT in healthcare involves the real-time acquisition of data from wearable sensors, which is subsequently transmitted to models for simulation and assessment. The history records are stored in the cloud to provide individualised healthcare treatments.

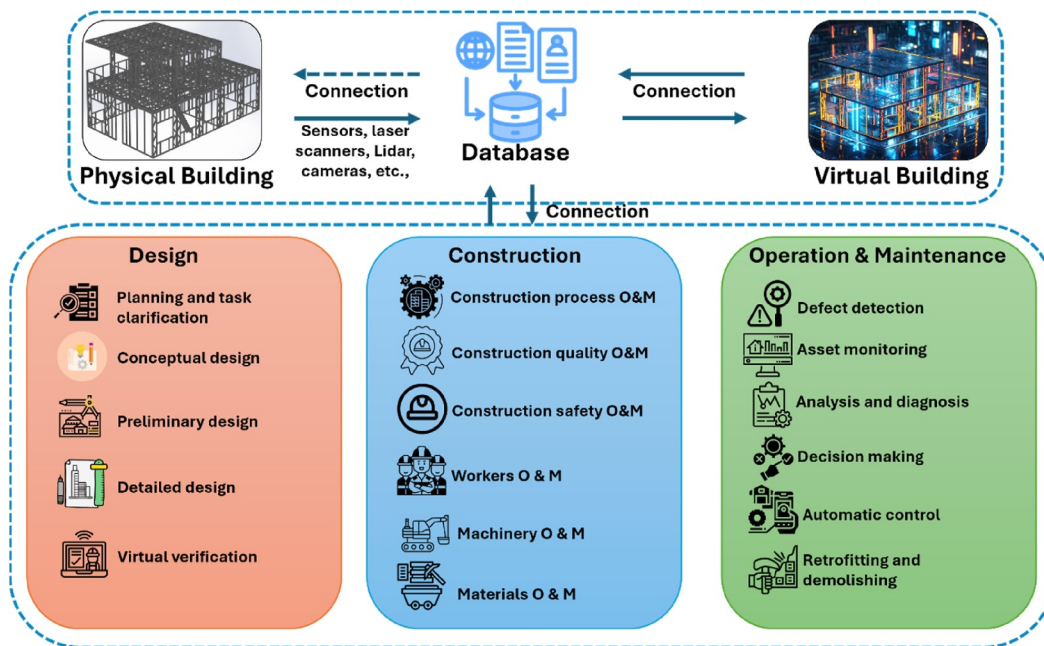


FIGURE 16 | Construction DT.

In this situation, the virtual counterpart of the actual building mirrors it. Information from the construction is gathered using several sensors including laser scanners, lidar, cameras and others. These data entering the DT will enable us to give a current and accurate picture. From original design and construction to ongoing operation and maintenance, every phase of a building's lifecycle employs the DT. It is used in the design phase for virtual verification and problem detection even before building starts. Over the course of a building project, this application might be used to control safety, guarantee quality and track development. The DT tracks assets, detects problems and uses data analysis and decision-making after the building is

finished. DTs offer potential to make building projects ecologically friendly, resilient to setbacks and efficient [76, 77].

5.8 | Telecommunication

Digital transformation is being used in many industries, including telecoms. In this sense, telecommunications refer to the communication networks and signals connected with the Internet of Things (IoT) used in any technical setting. IoT developments have improved the function of information

technology (IT) in modern businesses by transforming IT from only providing surveillance services to building a whole information framework for industrial operations. Smart services offered by many linked gadgets include environmental monitoring, traffic control, assisted living and building surveillance. Still, there are many difficulties in the connection between smart devices and systems including privacy issues, expensive limited bandwidth and network transmission delays. Moreover, the cost of high-performance cloud servers makes cloud-based solutions for big-scale industrial videography unworkable.

DT was used to improve communication between smart connected devices within industrial IoT by means of a cross-technology communication platform functioning in real-time deep learning directly on end devices, therefore addressing these difficulties. CTC enables the information flow among heterogeneous devices, thus immediately and directly meeting the industrial IoT needs. Figure 17 shows the CTC platform's architectural style [9].

6 | Challenges

The use of DT technology poses many difficulties depending on the degree of the integration complexity and size. Review of

current literature reveals five basic issues that are common in many fields, therefore addressing both the research sub quest (SQ1) and the more general main research question (RQ). Managing data trust, privacy, cybersecurity, convergence, governance, acquisition, and large-scale analysis is one major problem. Particularly in fields like social disputes, sociopolitical challenges, social inequity, and environmental sustainability, designers may struggle to duplicate behaviours that cannot be measured. These difficulties especially apply at lower degrees of System Readiness Level (SRL), when knowledge of the effects on society and stakeholders is vital. Moreover, they connect to maturity levels 3 and 4 in which filling digital models with real-time information becomes a crucial barrier in complicated DT implementations [11, 78].

Lack of Standards and Frameworks: DT implementations suffer in lack of accepted norms, frameworks and laws. Studies abound showing how this lack of interoperability particularly affects the industrial sector. Fostering a shared knowledge and enabling more general acceptance depends on research clarifying ideas, advantages and DT designs. Surveys and literature reviews may assist to address this difficulty thereby improving fundamental information of DTs. The financial load of implementing DTs is significant as sophisticated computing resources and many sensors are required. This great expense restricts

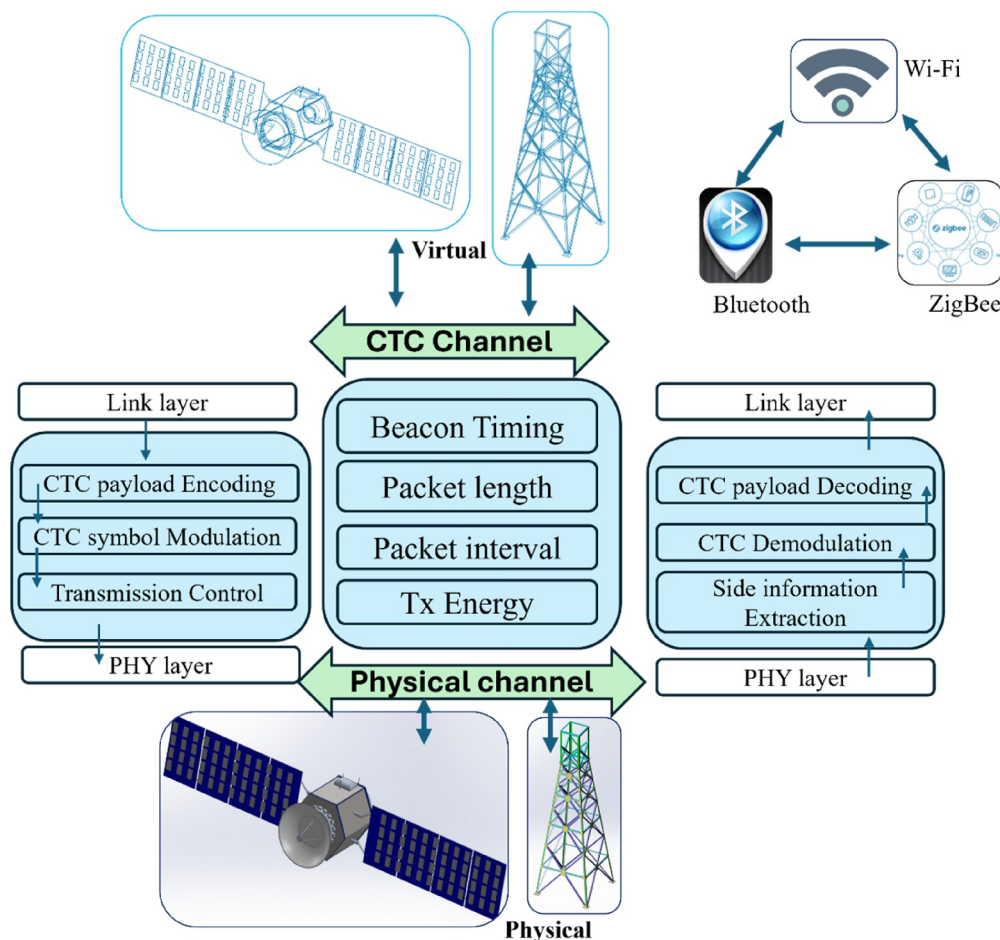


FIGURE 17 | The architecture of CTC.

accessibility especially in underdeveloped nations where resources could be limited. The complexity brought about by the growing number of sensors also complicates data transmission and processing, thereby making it challenging to reach maturity level 3 in the spectrum of development. This cost difficulty limits practitioners from attaining higher technology readiness levels (TRLs), in which case pilot systems may be clearly shown. Successful DT implementations depend on using artificial intelligence and big data to satisfy high long-term data analysis needs. The enormous volumes of created data need for strong algorithms and IoT technologies to enable efficient analysis. Still, the flow of data from many indicator systems hampers the creation of consistent regulations and standards [79, 80].

By means of real-time information transmission and perhaps autonomous operations, addressing this difficulty might improve levels 4 and 5 of maturity. Enhanced communication networks are desperately needed to enable DT applications in their whole. Improving communication among many sensors and equipment depends on technologies, such as 5G, which also provide low power consumption and fast access with better dependability. In DT settings, real-time data communication and operational efficiency depend on such developments. These difficulties draw attention to important areas requiring focus for effective use of DT technology in many spheres. Dealing with these problems would not only improve the general success of digital transformation projects but also help to enable better integrations.

6.1 | DT Challenges in India

RQ5. *What are the problems with using a DT-based smart city tool in India and other places?*

Using a DT-based smart city platform in India and other countries comes with many challenges including scientific, infrastructure, economical and governance ones. One may classify these limitations into the following main spheres: DT relies much on real-time data for correct modelling, hence data collecting and management is very important. Underdeveloped locations in India may lack the tools needed to gather and arrange high-quality real-time data. Data silos and inadequate integration of present technologies might create gaps in the digital representation of the smart city. Dealing with enormous volumes of municipal and personal data creates privacy and security issues. Globally, especially considering growing hacking and surveillance concerns, safeguarding private information is a key concern. Building a comprehensive DT for a large-scale smart city is challenging and requires the integration of multiple systems and technologies due to technical complexity. The compatibility with sensors, IoT devices, software platforms and older systems might be challenging in India and foreign countries. The real-time modelling and monitoring components of DT need algorithms and strong computer resources, which might be challenging to implement in societies without a mature IT infrastructure [81].

Digital transformation platforms depend critically on consistent, ubiquitous internet connection, especially 5G and IoT

infrastructure. Rural regions and even certain Indian cities might have sluggish or uneven internet connection, therefore restricting the compiling and distribution of real-time data. IoT sensors must be physically placed all around cities, which calls for significant financial and planning preparation. Urban surroundings might not always be ready for the large sensor deployment required by a DT-based platform. Building a smart city platform using DT technology requires not just financial but also major technical, infrastructure and professional labour costs. For nations, such as India, that are still growing, the financial weight might be rather burdensome. Budget restrictions could restrict the scope of such initiatives even in rich countries. The benefits of a DT-based smart city platform could not be evident for years, so it will be challenging to justify these long-term expenditures without rapid returns for societies with limited means [82].

Policy and Governance Concerns: Lack of defined policies and uniform standards limits the use of DT technology in smart cities, leading to unequal development and interoperability problems across platforms, therefore complicating worldwide or even national integration. Governments must negotiate difficult legal systems controlling data ownership, privacy and accountability. Standards for data governance are still under development in India and many other nations, which begs questions around who owns and manages data on a DT platform. Establishing and administering a DT-based smart city platform calls for personnel skilled in artificial intelligence, big data, IoT, urban planning. In many countries, such as India, professionals with the technical knowledge required to manage such sophisticated systems are rare. The degree of public involvement and knowledge will determine how well a smart city project succeeds. Although privacy issues may lower public adoption globally, low degrees of digital literacy in India might limit citizen participation in smart city services. Variations in Indian technology infrastructure across urban and rural regions might restrict the broad acceptance of smart city concepts. Regarding the development of smart cities, rural regions and smaller towns are often neglected all throughout the world. Social willingness to embrace technology-based governance varies in degree. Particularly in many locations with a digital divide, people might object to changes in how they use municipal services.

7 | Conclusion

A new age in data-driven decision-making, enhanced operational efficiency and predictive maintenance has begun with the widespread use of DT technology in fields as diverse as smart cities, manufacturing, healthcare and many more. DT offers real-time insights, modelling, process optimisation and a real-time virtual replica of physical system capabilities. However, its widespread use presents significant challenges, particularly in terms of data availability, infrastructure, scalability and security. Adoption of DT platforms is further difficult in developing countries, such as India, by constraints such as inadequate connectivity, high implementation costs and a scarcity of competent personnel. Assuring standards globally, fixing privacy concerns and getting past budgetary constraints are the three main obstacles to DT's full potential. Despite these obstacles, DT technology can significantly improve system

monitoring, industrial production and urban planning. As IoT, artificial intelligence and big data analytics develop, DTs' role in encouraging innovation and sustainability will grow, affecting the future of smart cities and many other businesses. This paper underlines the critical need of addressing technological, financial and governance barriers to effectively using DT platforms in both developed and developing countries. Although DT has a promising future, its full potential can only be realised via collaborative efforts in policymaking, infrastructure investment and technical knowledge development.

7.1 | Future Scope

RQ6. *What are the future trends in how people will use technology that makes things possible?*

The direction of the future of this study will be to understand the role that new technologies will play in innovations and human-technology interactions in the future. With the shift to Industry 5.0, the adoption of the so-called digital twin (DT), artificial intelligence (AI) and the Internet of Things (IoT) will become central to promoting automation and efficiency and enhancing sustainability. By exploring the development of these technologies to improve real-time decision-making, predictive maintenance and adaptive system design, future research may explore the application of arenas including manufacturing, healthcare, transportation and smart cities. Furthermore, it will be crucial to deal with the issues of data privacy, ethical use and human adaptability as well as to make the implementation responsible. The focus should also be on creating human-centric digital ecosystems, which will help to encourage collaboration between humans and intelligent systems. Finally, the future studies should seek to establish an inclusive, ethical and sustainable technology framework that can lead to innovation and the long-term vision of the industry 5.0.

Author Contributions

Bhargav Prajwal Pathri: conceptualization, writing – review and editing, supervision. **Krishna Vamshi Ganduri:** methodology, formal analysis.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon request. The data are not publicly available due to privacy or ethical restrictions.

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